MODULE 10 · SECTION 2 - INGESTING AND PROCESSING THE DATA

Planning Pipelines

Sources, sinks, the right service for the job, and the network the workers run in.

SECTION 2 of 5 EXAM WEIGHT 25% MODULE 10 of 45

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Before you pick a service, answer three questions: **how fresh, how complex, and who maintains it.**

Batch, micro-batch or streaming

PATTERN	FRESHNESS	AT LARKSPUR
Batch	Hours, on a schedule	Nightly billing and settlement
Micro-batch	Minutes, small chunks	Regional demand for Westgrid
Streaming	Seconds, record by record	Live readings in the customer app

THE TIEBREAKER

Choose the slowest pattern the business can accept. Streaming costs more to run and to reason about.

ETL or ELT?

ETL

Transform, then load

- Clean in Dataflow, Spark or Data Fusion
- Only finished data lands
- Right when raw data must not land
- Or when logic is not SQL

ELT

Load, then transform

- Land raw data in BigQuery
- Transform with SQL and Dataform
- Raw data kept for reprocessing
- The usual default on BigQuery

THE TIEBREAKER

ELT is the default. ETL wins when data must be masked before it lands, or the logic needs real code.

Who should do the work?

SERVICE	SWEET SPOT	SIGNAL IN THE QUESTION
Dataflow	Batch and streaming in one model, autoscaled	Streaming, late data, complex code
Managed Service for Apache Spark	Existing Spark and Hadoop code	Spark, Hive, lift and shift
Cloud Data Fusion	Visual pipelines, many connectors	Low-code, GUI, many sources
Dataform	SQL transformations inside BigQuery	SQL, ELT, version control, tests
BigQuery SQL	Scheduled and continuous queries	Data already in BigQuery
Cloud Run functions	Small reactions to single events	One file arrives, call one thing

THE TIEBREAKER

Match the service to the people and the code that already exist, not to the size of the data.

A quick decision path

01

Already in BigQuery?

SQL can do it: Dataform or scheduled queries.

02

Spark code exists?

Keep it: Managed Service for Apache Spark.

03

Low-code team?

Visual and connectors: Cloud Data Fusion.

04

Streaming or complex?

Code at scale, late data: Dataflow.

05

One small event?

A few seconds of glue: Cloud Run functions.

HOW TO USE IT

Stop at the first question that says yes. It is a starting point, not a law.

Names and newcomers

- **Cloud Functions** is now **Cloud Run functions**. The original is **1st gen**.
- Event-driven functions run up to **9 minutes**. HTTP functions up to **60 minutes**.
- **Cloud Data Fusion** keeps its name, and runs on short-lived **Spark** clusters it creates.
- **BigQuery continuous queries**: SQL that never stops, writing to Pub/Sub, Bigtable or tables.

> REMEMBER
>
> Older questions say Cloud Functions. Same product family, answer the same way.

Continuous queries or Dataflow?

CONTINUOUS QUERIES

Streaming in SQL

- Data already flowing into BigQuery
- Needs an Enterprise reservation
- Joins and windows still Preview
- Reprocessing can duplicate output

DATAFLOW STREAMING

Streaming in code

- Any source, any sink
- Watermarks and late data
- Exactly-once processing options
- More to build and operate

THE TIEBREAKER

Simple SQL on data already in BigQuery: continuous queries. Late data, state or many sinks: Dataflow.

Larkspur's pipelines, end to end

SOURCE	DOES THE WORK	SINK
Pub/Sub meter-readings	Dataflow streaming	raw.meter_readings, Bigtable meter_live
Oracle billing	Datastream	BigQuery raw
Westgrid files in S3	Storage Transfer Service	Cloud Storage landing bucket
raw and curated tables	Dataform	curated and marts

THE PATTERN

One stream can feed two sinks: history for analysts, the latest value for the app.

Where Dataflow workers live

- Workers are VMs in **your subnet**. Size its IP range for autoscaling.
- **No public IPs**? Then turn on **Private Google Access** to reach Google APIs.
- Need PyPI or another outside site? **Cloud NAT**, or a **custom container**.
- Without Streaming Engine or Shuffle, workers talk on **TCP 12345 and 12346**.

> REMEMBER
>
> Behind VPC Service Controls, workers must use restricted.googleapis.com.

Launching Larkspur's pipeline, privately

```
1  SUBNET="https://www.googleapis.com/compute/v1/projects/lk-network-prod/\
2  regions/europe-west2/subnetworks/lk-dataflow"
3
4  python -m larkspur.readings_pipeline \
5    --runner=DataflowRunner \
6    --project=lk-ingest-prod \
7    --region=europe-west2 \
8    --subnetwork="$SUBNET" \
9    --no_use_public_ips \
10   --service_account_email="$LK_WORKER_SA" \
11   --dataflow_kms_key="$LK_DATAFLOW_KEY" \
12   --enable_streaming_engine
```

1-2 A Shared VPC subnet in the host project needs its full URL.

9 No public IPs, so the subnet needs Private Google Access.

10-11 A dedicated worker identity, and Larkspur's own key.

A BONUS

Streaming Engine also removes the need for worker-to-worker firewall ports.

Two identities, one key

- The **Dataflow service agent** creates workers and checks quota. Leave its roles alone.
- The **worker service account** reads and writes your data. Make it dedicated and minimal.
- **Shared VPC**: grant **Compute Network User** in the host project to the Dataflow service account.
- Data is encrypted in transit and at rest by default. A **CMEK** covers disks and state.

> THE SIGNAL
>
> 'Permission denied on the subnet' is a Network User problem, not a BigQuery role problem.

Phrases that pick the answer

THE QUESTION SAYS	REACH FOR
SQL team, data in BigQuery, version control	Dataform
Existing Spark jobs	Managed Service for Apache Spark
Visual, low-code, many connectors	Cloud Data Fusion
Streaming, late data, exactly-once	Dataflow
Must not land unmasked	ETL: transform before loading
No external IPs on workers	Private Google Access on the subnet

THE PATTERN

Freshness, complexity, and the people who will maintain it.

> Larkspur's analysts own the SQL that turns raw readings into curated tables and marts. Today it lives in saved queries that run in the wrong order on bad days.
>
> They want version control, tests, and steps that run in dependency order. Nobody on the team writes Java or Python.

What should they use?

A. A Dataflow batch pipeline, rewritten in Java, that runs once a night

B. Dataform, with each step as SQLX and assertions checking the results

C. A Cloud Data Fusion pipeline that copies tables between datasets

D. A Cloud Run function that runs the saved queries one after another

◆ PAUSE AND DECIDE

What should they use?

A **Dataflow in Java**

Nobody can maintain Java, and the data is already in BigQuery.

B **Dataform**

SQL they already know, in Git, with tests and dependency order built in.

C **Cloud Data Fusion**

Moves data through Spark clusters for work BigQuery can do itself.

D **A Cloud Run function**

Hand-written ordering, no tests, and a 9-minute limit on events.

THE SIGNAL

SQL team, data in BigQuery, version control and tests: that is Dataform's whole job.

> An organisation policy forbids external IP addresses on every VM. Larkspur's new Dataflow job starts, but its workers cannot reach Pub/Sub or BigQuery.
>
> Security also says nothing in the pipeline may reach the public internet.

What should the team configure?

A. Give the worker service account the BigQuery Admin and Pub/Sub Admin roles

B. Add a firewall rule allowing TCP 12345 and 12346 between the workers

C. Ask for a policy exception, so that the workers can have external IPs

D. Run with no public IPs, and turn on Private Google Access for the subnet

◆ PAUSE AND DECIDE

What should the team configure?

A Admin roles

Roles do not create a network path, and Admin breaks least privilege.

B Worker firewall ports

Those ports are for workers talking to each other, not to Google APIs.

C A policy exception

It reopens the internet path that security has ruled out.

D Private Google Access

Internal-only workers reach Google APIs, and nothing reaches the internet.

THE SIGNAL

'No external IPs' plus 'cannot reach Google APIs' means Private Google Access on the subnet.

> Larkspur's networks live in a Shared VPC host project, lk-network-prod. A Dataflow job in the service project lk-ingest-prod must run in its subnet.
>
> The job fails at launch with a permission error on the subnetwork.

What fixes the launch?

A. Give the Dataflow service account Compute Network User on the host subnet

B. Give the worker service account the Dataflow Admin role in the host project

C. Create a VPC in the service project, and peer it with the host network

D. Give the engineer who launches the job the Owner role on the host project

◆ PAUSE AND DECIDE

What fixes the launch?

A Network User for Dataflow

The documented grant, with the subnet's full URL in the job.

B Dataflow Admin for workers

Wrong identity and wrong role. Workers do not attach the subnet.

C A peered VPC

Sidesteps the shared network that the platform team runs.

D Owner for the engineer

Far too broad, and Dataflow's own account still lacks access.

THE SIGNAL

A permission error on a Shared VPC subnet is a Compute Network User grant in the host project.

> Larkspur Energie receives German customer files each night. Each record holds a national ID number that must be tokenised.
>
> Security says the raw IDs must never be stored in BigQuery, not even in a raw dataset for a few minutes.

How should the pipeline be designed?

A Load the files into a raw dataset, and have Dataform tokenise into curated

B Load the files into BigQuery, and protect the ID column with a policy tag

C Tokenise in a Dataflow pipeline, and write only tokens into BigQuery

D Load with Data Fusion, then apply dynamic masking to the ID column

◆ PAUSE AND DECIDE

How should the pipeline be designed?

A Dataform after loading

Classic ELT, so the raw IDs land in BigQuery first.

B A policy tag

Hidden from most users, but the raw IDs are still stored.

C Tokenise in Dataflow

ETL: the IDs are replaced in flight, so only tokens ever land.

D Dynamic masking

Masking changes what people see, not what is stored.

THE SIGNAL

'Must never be stored, not even raw' rules out every load-then-fix design. That is ETL.

> The customer app shows each meter's latest reading. Readings arrive on Pub/Sub at up to 10,000 a second in the evening, including resent duplicates.
>
> The app must see a new reading in Bigtable within seconds, and the pipeline must scale on its own.

Which service should process the stream?

A. A Dataflow streaming pipeline from Pub/Sub, writing to Bigtable

B. A Cloud Run function per message, writing each one to Bigtable

C. A scheduled query every minute that copies new rows into Bigtable

D. A Spark batch job every five minutes on a long-running cluster

◆ PAUSE AND DECIDE

Which service should process the stream?

A Dataflow streaming

Seconds of latency, autoscaling, and state to drop duplicate resends.

B A function per message

It scales, but each call is alone, with no shared state for duplicates.

C A scheduled query

A minute behind at best, and it needs the data in BigQuery first.

D Spark every five minutes

Minutes, not seconds, and a cluster to keep running.

THE SIGNAL

Seconds, scale on its own, and duplicates to handle: a streaming pipeline with state.

> When a Westgrid settlement file lands in Larkspur's landing bucket, the team wants to post a message to the operations channel and start the load workflow.
>
> Files arrive a few times a day. The work takes a couple of seconds, and nobody wants a server to look after.

What should react to each new file?

A. A Dataflow streaming pipeline that watches the bucket around the clock

B. A Cloud Run function, triggered by the bucket's object finalised event

C. A Cloud Data Fusion pipeline, scheduled to run once every minute

D. A cron job on a small Compute Engine VM that lists the bucket

◆ PAUSE AND DECIDE

What should react to each new file?

A Dataflow, always on

Workers running all day for a few seconds of work.

B A Cloud Run function

Runs only when a file lands, for seconds, with no server to manage.

C Data Fusion every minute

A Spark cluster spun up every minute to post a message.

D Cron on a VM

A server to patch and pay for, and polling instead of events.

THE SIGNAL

One event, a few seconds of work, no server: Cloud Run functions.

Module 10 in five lines

- Ask three questions: **how fresh, how complex, who maintains it**.
- **ELT** by default on BigQuery. **ETL** when raw data must never land, or logic needs code.
- **Dataform** for SQL, **Spark** for Spark, **Data Fusion** for low-code, **Dataflow** for streaming.
- **Cloud Run functions** for one small event. **Continuous queries** for SQL streaming.
- No public IPs needs **Private Google Access**. Shared VPC needs **Network User**.

COMING UP

Next: module 11, Pub/Sub delivery, replay and failure.